EXPLORING THE FUTURE OF TECHNOLOGY

Drones and UAS (Unmanned Aerial Systems)

PRESENTED BY SIYA SHARMA

Growth Timeline

Market Valuation of USD 2.7B

India's drone market reached **USD 2.7B** in 2022, marking significant initial growth.

Rapid Startup Growth

Numerous startups emerged, driving **innovation** and expanding the **UAS** ecosystem across India.

Government Investments Announced

The government introduced **initiatives** fueling market growth, totaling **INR 120 crore** in funding.

Funding Surge in 2023

Drone funding tripled in 2023, with **over $50M** raised from various investors.

Projected Market Expansion

The market is expected to **grow to USD 13B** by 2030, showcasing immense potential.

Major Contributors to India's Drone Revolution

Key Players

IdeaForge is a leading drone manufacturer in India, known for its innovative solutions and eyes on an upcoming IPO in 2023.

Garuda Aerospace

Garuda Aerospace has gained recognition for its **advanced agricultural drones**, significantly enhancing productivity and efficiency in the farming sector across India.

Asteria Aerospace

Asteria Aerospace, associated with Reliance, develops cutting-edge drones for various applications, including surveillance and logistics, positioning itself as a key player.

Investment Surge

In 2023, **funding tripled**, highlighting growing investor confidence in India's drone market as startups innovate across various sectors.

Major Investors

Key players like Qualcomm Ventures and InfoEdge have fueled the **drone ecosystem**, investing over $50M+ to support emerging technologies.

Funding Landscape

The funding environment remains robust, attracting diverse **investments** that pave the way for new developments and market expansion.

Investment Trends and Landscape Insights

Government Support Strategies



Drone Shakti

The Drone Shakti initiative aims to promote **indigenous manufacturing** and innovation in the drone industry.

Kisan Drone Scheme

This scheme provides farmers with access to drones, enhancing **agriculture efficiency** and promoting technological adoption.

Drone Rules Overview

The **Drone Rules 2021** established a framework for the safe operation of drones, ensuring compliance and security for operators and users across India.

Digital Sky Platform

The **Digital Sky Platform** is a centralized system for drone registration, flight approvals, and real-time tracking, enhancing operational transparency and efficiency.

NPNT Protocol

The **No Permission, No Takeoff (NPNT)** protocol ensures that drones comply with regulatory requirements before flight, promoting safety and responsible usage in the airspace.

Navigating India's Regulatory Landscape for Drones

Challenges in Drone Sector



Data Privacy

Ensuring **user data protection** is essential for gaining public trust and compliance with regulations.

Airspace Management

Coordinating drone flights within shared airspace is vital to prevent accidents and ensure safety.

Defence Applications

Drones enhance **surveillance and reconnaissance**, providing vital information for military operations while minimizing risks to personnel in conflict zones.

Agricultural Innovation

In agriculture, drones offer **precision farming** solutions, enabling farmers to monitor crops, optimize resource use, and increase overall productivity efficiently.

Logistics and Delivery

Drones streamline **last-mile delivery**, reducing transportation costs and time, while enhancing supply chain efficiency in urban and rural areas.

Exploring Opportunities for Drones Across Industries

Urban Air Mobility

Urban air mobility will revolutionize **transportation efficiency** by enabling drone taxis and delivery services in congested cities, addressing traffic issues effectively.

Sustainable Drones

The development of **sustainable drones** focuses on eco-friendly materials and energy sources, reducing emissions and promoting environmental conservation in drone operations.

Training and Education

Investing in training and education will ensure a skilled workforce, fostering **innovation** and safety standards necessary for the growing drone sector in India.

Innovations and Developments in Drones

“India’s drone sector is on the brink of revolutionizing multiple industries, setting new standards in innovation worldwide.”

— Dr. Ramesh K. Singh, Aerospace Industry Expert

Thank You.

Presented by Siya Sharma